System Validation Report - Online Internship & Placement Preparation Portal

Project	Online Internship & Placement Preparation Portal (OIPPP)
Release	1.0 (tag `v1.0.0`) plus the fixes listed in section 8
Test window	2–6 Nov 2026
Prepared by	Guru R Bharadwaj (QA lead), with results from all four members
Date	07-11-2026 (final)

1. Scope

This report validates release 1.0 of OIPPP against the SRS, using the Software Test Plan v1.0 approach and the test cases in the Test Case Document v2.0. It covers:

• system tests of every test case on the deployed test build, through the API and the UI;

• the full Postman collection on the deployed build;

• security validation (Test Plan 5.1) on the deployed hostnames;

• the performance rerun on the deployed build (PERF-1 to PERF-3);

• user acceptance testing (QA-1) and the device check (QA-5).

Unit and integration test results are in the Unit Test Report. Coverage is in the Code Coverage and Branch Coverage reports.

2. Build and environment under test

Item	Value
Client	https://oippp-client.onrender.com (Render static site, Singapore)
API	https://oippp-api.onrender.com/api (Render free web service, Singapore, Node 22)
Database	MongoDB Atlas M0 replica set (Mumbai), database `oippp`
Email	Mailtrap sandbox inbox
Build commit	`f044d21` (v1.0.0 plus the fix for #99) for the 2 Nov runs in section 4; `main` from 4 Nov, after PR #100 was merged. Final: `dev`/`main` @ `7ab69d2` plus this report; remaining identity, listings UI and applications UI cases run 06-11-2026
Data	`seed:demo`: 1 admin, 10 recruiters, 200 students, 5,000 listings, 100 questions, 5 tests, 600 applications
Details	deployment.md section 6

3. Entry and exit criteria (Test Plan section 5)

Criterion	Status	Evidence
Entry: stable build deployed to the test environment	Met, 02-11-2026	Post-deploy checks in deployment.md section 6
Entry: seed data loaded	Met, 02-11-2026	`seed` and `seed:demo` output
Entry: test cases reviewed	Met, 05-11-2026	Identity (#83, PR #105), listings (#87, PR #109), applications (#91, PR #101) and aptitude and analytics (#95, PR #114) merged
Exit: all planned test cases executed	Met with exceptions	All 51 cases executed (section 4.1): 39 pass, 4 partial, 2 fail, 6 blocked, 0 not run. Blocked cases need the admin account or database access on Render (5 identity) or a 2,000-student dataset (PERF-3); all pass in the automated suite

Criterion	Status	Evidence
Exit: no open Critical or High defects	Met, 07-11-2026	Section 8: #99 fixed; #118 Low, #123 Medium and #124 Medium are recorded as deviations; ZAP found 0 High
Exit: server statement coverage at least 70% (QA-4)	Met	96.16% statements, 94.37% branches on the final commit (Branch Coverage Report section 7)
Exit: every requirement in the RTM has at least one passed test	Met for automated tests; partly met on the deployed build	Every requirement has a passing automated test (Unit Test Report). Section 5 lists the requirements without a deployed-build pass

4. Test case results

4.1 Summary

Module	Owner	Test cases	Pass	Partial	Fail	Blocked	Not run
Identity and security	Deekshith Gowda G (#84)	14	7	1	1	5	0
Listings and search	Sharath H G (#88)	11	8	2	0	1	0
Applications, review and notifications	Guru R Bharadwaj	9	9 (API; UI also for TC-APP-01, TC-APP-03, TC-REV-01)	0	0	0	0
Aptitude and analytics	E S Sukruth (#96)	17	15 (3 Render, 12 local)	1	1	0	0
Total		**51**	**39**	**4**	**2**	**6**	**0**

Counting rule: a case counts as Pass only if every step run passed; TC-AUTH-04 is Partial because its admin path was not run, and TC-PERF-01 is Partial because PERF-1 passes and PERF-2 fails (#123). Applications UI: TC-APP-02 and TC-REV-02 UI steps were not run and TC-REV-05 UI is partial (badge and list checked, Mark read not pressed); their API steps pass. Identity also ran 12 login, RBAC and injection checks on the deployed build (section 4.3); they are not counted because they are not separate test cases in identity.md.

4.2 Applications, review and notifications

API steps of the nine applications test cases, run with a script against `https://oippp-api.onrender.com/api` on 02-11-2026, build `f044d21`. Users R1, R2, S1 and S2 are `recruiter01`, `recruiter02`, `student0001` and `student0002`. R1 created listings L1 to L3 for the run. Each check compares the status code, error code, message and stored data with the expected result, and email checks read the Mailtrap inbox through its API.

Result (second run, 13:57–14:02 UTC): all nine test cases pass their API steps. The script ran 50 checks (3 setup, 47 test) and 49 passed in the run. The 50th, the TC-REV-03 rejection email, arrived after the script's 2-minute wait, through the app's retry (see below). UI steps, run on 06-11-2026 with Playwright/Edge: TC-APP-01, TC-APP-03 and TC-REV-01 pass; TC-REV-05 partial (badge and list checked, Mark read not pressed); TC-APP-02 and TC-REV-02 UI not run.

Test case	API checks	Result	Notes
TC-APP-01	7	Pass	One application, one history entry, resume copy opens; 1,001-character note gives 400
TC-APP-02	4	Pass	409 `LISTING_CLOSED`, 422 `NOT_ELIGIBLE` naming 8.0, 403 for a recruiter, nothing stored
TC-APP-03	4	Pass	Newest first, history newest first, S2 sees only S2's applications
TC-APP-04	2	Pass	CGPA 7.99 gives 422; 8.0 gives 201

Test case	API checks	Result	Notes
TC-REV-01	6	Pass	Shortlist email in Mailtrap 6 s after the change, same subject as the notification
TC-REV-02	6	Pass	Page shape, status filter, limit 51 gives 400, owner gets the PDF, R2 gets 403 `NOT_LISTING_OWNER`
TC-REV-03	11	Pass	Rejection email ends with "Reason: Skills do not match the role."; it arrived on the third attempt, 14:06:32 UTC
TC-REV-04	2	Pass	Two simultaneous shortlists: one 200, one 422 `INVALID_TRANSITION`; one history entry, one notification
TC-REV-05	5	Pass	Unread list, mark read, other user's notification 404, no token 401

First run (13:47–13:53 UTC): 48 of 50, both email checks failed. No email reached Mailtrap, yet every notification was stored with email status `SENT`. The API instance was running without the SMTP settings, so the mailer fell back to its no-send JSON transport, which reports success. The settings were present on the service, and a fresh deploy of the same commit (live at 13:56 UTC) picked them up. A one-off check after the deploy received its email in 6 s. No code change was needed. See observation O-1 in section 8.

Email retry seen on the deployed build (REV-4). Mailtrap's free sandbox refuses emails sent in quick succession (`550 5.7.0 Too many emails per second`). TC-REV-03 changes status three times in a few seconds, so two of its emails were refused:

Email	Attempt 1	Retry	Delivered
"QA L1: Offer" to S1	13:58:51, refused	after 1 minute	14:00:32, attempt 2
"QA L1: Rejected" to S2	13:58:56, refused	after 1 and 5 minutes	14:06:32, attempt 3

The status changes themselves returned 200 at once, so email failures did not fail the request. This matches REV-4's retry schedule of 1, 5 and 15 minutes. On a paid email provider the refusals would not happen. For later runs, wait a few seconds between status changes when checking email.

4.3 Identity and security

Issue #84 checks (below) ran on 04-11-2026; the remaining identity.md cases ran on 06-11-2026 with API scripts and Playwright/Edge (`evidence/identity-render-run*.mjs`, `ui-register-run.mjs`).

Test case	Result	Notes
TC-AUTH-01 Duplicate email	Pass	409 `DUPLICATE`
TC-AUTH-02 Password validation	Pass	Short, no-digit and no-letter passwords give 400
TC-AUTH-03 Account lockout	Pass	6th attempt 423 `ACCOUNT_LOCKED` with `retryAt`; login works after the lock ends
TC-AUTH-04 Resume download authorization	Partial	Owner 200 PDF, other student 403, other recruiter 403, no token 401; admin path not run
TC-AUTH-05 JWT expiry	Pass	Local unit test (60 min)
TC-AUTH-06 Student registration	Pass	API 201 and UI
TC-PROF-01 Resume upload validation	Pass	20 KB PDF 200; 2.1 MB 413; `.docx` rejected
PROF-1 Profile editing	Pass	Deployed UI
TC-SEC-01 bcrypt hashing	Blocked	Needs database access; passes in the automated suite

Test case	Result	Notes
TC-SEC-02 Login rate limit	**Fail**	11 bad logins in a minute all 401, no 429; `ratelimit-remaining` jumps between buckets (bug #124, DEVIATIONS row 30). Passed on 02-11 (section 4.6) and in the automated suite
PROF-3 Recruiter approval	Blocked	Admin credentials not available to testers
PROF-4 Deactivation revokes tokens	Blocked	Admin credentials; irreversible on shared data
DOM-2 One-offer policy	Blocked	Policy is off on Render; enabling it needs admin. Apply after OFFER returned 201 as expected with the policy off
SEC-6 Audit log integrity	Blocked	Admin credentials

Identity summary: 7 Pass, 1 Partial, 1 Fail, 5 Blocked. The blocked cases all pass in the automated suite (`admin.test.js`, `audit.test.js`, `auth-service` unit tests).

Deployed UI checks

Check	Result
Student login	Pass
Student profile editing and persistence	Pass
Student access to Admin UI	Pass
Recruiter login	Pass
Recruiter access to Admin UI	Pass
Unauthenticated access to Admin UI	Pass

Deployed API checks

Check	Result
Unauthenticated GET /auth/me	Pass, HTTP 401
Student GET /admin/users	Pass, HTTP 403
Recruiter GET /admin/users	Pass, HTTP 403
Admin GET /admin/users	Pass, HTTP 200
`$ne` injection	Pass, HTTP 400 VALIDATION
`$gt` injection	Pass, HTTP 400 VALIDATION
`$where` injection	Pass, HTTP 400 VALIDATION
XSS login payload	API validation passed, HTTP 400; website rendering check not run

Local automated tests

Eight integration test suites were run locally: identity-flow, identity-edge-cases, RBAC, auth, profile, resume, admin and audit. Result: 109 tests passed, 0 failed. These are local automated results, not deployed system-test passes.

Details: test-cases/identity.md.

4.4 Listings and search

Issue #88 execution against the deployed test build on 04-11-2026. API checks used `https://oippp-api.onrender.com/api`. The exact deployed application commit SHA was not independently verified from Render Events.

Test case	Result	Evidence and outstanding work
TC-LST-01 — Listing creation, validation and expiry	Pass	Valid creation returned 201 and OPEN. Past deadline and negative stipend returned 400. A short-deadline listing eventually became CLOSED; editing it returned 409 `LISTING_CLOSED`.

Test case	Result	Evidence and outstanding work
LST-3 — Listing ownership	Pass	Other recruiter received 403 for edit and close. Owner successfully edited and closed the listing; GET confirmed CLOSED.
SRCH-4 — Saved internships and bookmark limit	Partial	Save returned 201; duplicate save returned the same bookmark ID; GET showed one occurrence; removal returned 204. Closed and expired listings returned 400 `LISTING_NOT_BOOKMARKABLE`. The 100/101 boundary and duplicate-at-limit case were not tested to avoid polluting shared seeded data.
SRCH-5 — Recommendations	Pass	1,455 recommendations returned. All were OPEN and not expired; match counts were consistent and ranking descending. Unauthenticated access returned 401; recruiter access returned 403. Empty profile (06-11-2026): `{"listings":[]}` and "No matches yet" in the UI.
TC-PERF-01 — Search/API load performance	Partial	50 users x 5 min on Render, 1,265 requests, 0% errors. PERF-1 search p95 1,370 ms: Pass. PERF-2 other calls p95 3,584 ms: Fail (bug #123, DEVIATIONS row 29). Cold start 32.5 s. performance-rerun.md
PERF-3 — Dashboard rendering	Blocked	Not rerun on Render: deployed seed has 200 students, not the 2,000 PERF-3 needs. Local 30-10-2026 run with 2,000 students: 0.19–0.29 s (Pass)
TC-SRCH-01 — Search, filters, sort and pagination	Pass	Search, mode, stipend and skill queries returned expected results. Stipend-descending order was monotonic across 50 records; pages 1 and 2 had no duplicate IDs. Invalid mode, sort and stipend range returned 400. Newman run: 10 requests, 15 assertions, 0 failures.
UI-LST-01 — Listing search page	Pass	Manual check showed 803 React search results. User confirmed filters, sorting and pagination worked.
UI-LST-02 — Recruiter listing form	Pass	06-11-2026, Playwright/Edge: empty submit shows field errors, negative stipend and past deadline rejected, owner edit saved, other recruiter cannot edit. Screenshots in `evidence/ui-render/`
UI-SRCH-04 — Saved internships UI	Pass	06-11-2026: save, "3 of 100 saved" count, single entry, removal
UI-SRCH-05 — Student home feed UI	Pass	06-11-2026: recommendation cards with skill chips, deadlines panel, search navigation, empty state

Listings summary: 8 Pass, 2 Partial (SRCH-4, TC-PERF-01), 1 Blocked (PERF-3), 0 Not run. PERF-2 inside TC-PERF-01 fails (#123).

Observation — public listing count: the public API initially reported 4,421 listings and later 4,427 after test listings were created. Seed documentation describes 4,500 open listings. The difference is unexplained and is recorded as an observation, not a confirmed defect.

The short-deadline listing was OPEN on an early poll and CLOSED on a later GET. Editing after the deadline returned 409 `LISTING_CLOSED`. The exact background-job closure time was not measured.

4.5 Aptitude and analytics

Issue #96, run on 05-11-2026 with `server/scripts/system-test-aptitude.js` (API, same calls as Postman) and by hand in a browser. Test cases: test-cases/aptitude-analytics.md. Evidence JSON: `evidence/`.

Two environments are used, and every result says which:

• **Render**: the Part-3 test build, `main` @ `1e07391` (the exact SHA was not re-checked on Render Events). Student and recruiter accounts only: the admin password is not in the repository, and deactivation cannot be undone on the shared data.

• **Local**: the same code (`dev` @ `8cb16a0`) on a team laptop: in-memory MongoDB replica set, `npm run seed` + `seed:demo` (the same data set as Render), Vite dev server, and the Chromium-based browser built into the Claude desktop app. Used for the admin cases, SAFE-2 and the UI steps. A local pass is not a Render pass.

Test case	Render	Local	Notes
TC-TST-01 Resume after refresh / closed tab	Pass (API)	Pass (UI)	Same attempt and `endsAt` after restart; 3 concurrent starts → one attempt
TC-TST-02 Auto-submit and score by topic	**Partial**	Pass	Local: sweep submitted 1.0 s after `endsAt`; 2 s before → 204, 5 s and 40 s after → 409 `ATTEMPT_CLOSED`; 4/6 by topic as expected. Render: hard cutoff and `TIMEOUT` scoring hold, but the sleeping free instance closed the attempt only on the next request, 8.5 h later (DEVIATIONS row 11)
TC-TST-03 Question bank	Role checks pass	Pass	Boundaries, stale version, archive
TC-TST-04 Draft, publish, archive	Role checks pass	Pass	`TEST_LOCKED`, `STALE_VERSION`, `INVALID_QUESTIONS`
TC-TST-05 Single attempt	Pass (API)	Pass (UI)	`ATTEMPT_EXISTS`, owner isolation, idempotent submit
TC-TST-06 Save within 1 s	**Fail** (#118)	Pass (max 29 ms)	Render: median 931 ms, p95 1,023 ms; `/health` alone median 694 ms. Recorded as DEVIATIONS row 28 (PR #119)
TC-ANL-01 Placement statistics and CSV	Role checks pass	Pass	CSV counts equal the filtered dashboard
TC-ANL-02 Student progress	Pass (API)	Pass (UI)	Empty state `averageScore: null` checked on Render
TC-DOM-03 Per graduation year	Not run	Pass	2027: 104, 2028: 96; 2099 empty; 1999 → 400
TC-SAFE-01 Refresh / close / offline in a browser	Not run	Pass (Chromium)	Offline simulated in the page; "Not saved" after 3 retries, saved again after reconnect; 360 px OK
TC-SAFE-02 Deactivation keeps data	Not run	Pass	Old token 401, login 403, totals unchanged (202 results, 600 applications)
UI-TST-01 Tests list	Not run	Pass	
UI-TST-02 Attempt screen and timer	Not run	Pass	Sticky timer, warning under 5 min, auto-move to result at 0:00
UI-TST-03 Result page	Not run	Pass	Matches the API
UI-TST-04 Question bank and test builder	Not run	Pass	Field messages, confirmations, Draft → Published → Archived
UI-ANL-01 Progress dashboard	Not run	Pass	Data in 234 ms
UI-ANL-02 Statistics dashboard	Not run	Pass	Data in 744 ms; filter, validation and CSV

Issue #96 summary (05-11-2026): 17 cases. Best result per case: 15 Pass (3 on Render, 12 on the local build only), 1 Partial (TC-TST-02 on Render), 1 Fail (TC-TST-06 on Render, bug #118). Firefox, Edge and a real Android phone are covered by #97.

4.6 Postman collection on the deployed build

OIPPP.postman_collection.json run with newman against `https://oippp-api.onrender.com/api` on 02-11-2026, 13:42–13:46 UTC, build `f044d21`:

```
npm run test:api -- --env-var baseUrl=https://oippp-api.onrender.com/api --env-var adminEmail=admin@oippp.dev --env-var adminPassword
```

Result: 112 requests, 157 assertions, 0 failures in 4 min 14 s. Average response time 561 ms (min 164 ms, max 2 s), measured from a team laptop.

Folder	Requests	Assertions	Failed
00 Setup	8	10	0
TC-AUTH-01	2	4	0
TC-AUTH-02	4	10	0
01 Second rate-limit window	5	5	0
TC-PROF-01	5	7	0
TC-LST-01	3	4	0
TC-SRCH-01	7	13	0
TC-APP-01	3	7	0
TC-AUTH-04	4	5	0
TC-APP-02	5	9	0
TC-REV-01	6	9	0
TC-TST-01	29	33	0
TC-TST-02	3	6	0
TC-ANL-01	3	8	0
DOM-2 one-offer policy	3	4	0
TC-AUTH-03	9	11	0
TC-LST-01 step 3	1	2	0
TC-SEC-02	12	12	0

Points this run proves on the deployed build that the local run could not:

• **TC-LST-01 step 3:** a listing with a 30-second deadline was closed by the server's background job on Render, and editing it then returned 409 `LISTING_CLOSED`.

• **TC-SEC-02 / SEC-5:** behind Render's proxy (`TRUST_PROXY=1`), the 11th login in a minute returned 429 `RATE_LIMITED`.

• **TC-TST-01 / TC-TST-02:** aptitude attempts use Atlas transactions, which need the replica set.

The run adds its own test users and listings and turns the one-offer policy back off at the end. It doesn't change the `seed:demo` data.

5. Requirement coverage

Every SRS requirement has a passing automated test (Unit Test Report; RTM in the Test Case Document). Deployed-build status from section 4:

Area	Requirements	Passed on the deployed build	Not passed on the deployed build
Identity	AUTH, PROF, RBAC	TC-AUTH-01 to 03, 05, 06, TC-PROF-01, PROF-1, RBAC and injection checks (section 4.3)	TC-SEC-02 fails (#124); TC-AUTH-04 admin path; PROF-3, PROF-4, DOM-2, SEC-6, TC-SEC-01 blocked (automated only)
Listings and search	LST, SRCH	TC-LST-01, LST-3, TC-SRCH-01, SRCH-5, PERF-1, all four UI cases	SRCH-4 partial (limit boundary); PERF-2 fails (#123); PERF-3 blocked
Applications and review	APP, REV, NOTIF	All nine cases, API steps (section 4.2); UI for TC-APP-01, TC-APP-03, TC-REV-01	UI steps of TC-APP-02, TC-REV-02 not run; TC-REV-05 UI partial
Aptitude and analytics	TST, ANL, DOM-3	TC-TST-01, TC-TST-05, TC-ANL-02	TST-4 (TC-TST-06, #118); TC-TST-02 partial; admin and UI cases passed locally only
Safety	SAFE-1, SAFE-2	SAFE-1 server side	SAFE-2 local only

Area	Requirements	Passed on the deployed build	Not passed on the deployed build
Security	SEC-1 to SEC-6	SEC-2, SEC-4; SEC-5 on 02-11 only	SEC-5 fails on the 06-11 rerun (#124); SEC-1, SEC-3, SEC-6 automated only
Quality	QA-1, QA-5	-	QA-1 (1 of 5 participants), QA-5 (device rows not run)

6. Non-functional results

Area	Requirement	Result	Source
Performance	PERF-1, PERF-2, PERF-3	Render 06-11-2026: search p95 1,370 ms (Pass), other APIs p95 3,584 ms (Fail, #123), PERF-3 not rerun. Local 30-10-2026 baseline: search p95 37 ms, other APIs p95 62 ms at 50 VUs; dashboard render 0.19–0.29 s for 2,000 students	#89 (performance-rerun.md), #88
Security	SEC-1 to SEC-6, ZAP baseline, npm audit	ZAP baseline on the client URL: 0 High, 2 Medium (CSP, SRI), 5 Low. `npm audit --omit=dev --audit-level=high`: 0 on server and client. SEC-1, SEC-3 to SEC-6 pass in the automated suite; SEC-5 passed on the deployed build on 02-11 (section 4.6) but failed the 06-11 rerun (#124: no 429 behind the Render proxy; account lockout still works); SEC-4 injection checks (`$ne`, `$gt`, `$where`) also pass on the deployed build (section 4.3). SEC-2 passes: on 06-11-2026 both `https://oippp-client.onrender.com` and `https://oippp-api.onrender.com` answered over TLS with a valid certificate (curl, no `-k`), plain HTTP returns 301 to HTTPS, and both send `Strict-Transport-Security` (client `max-age=315360000; includeSubdomains; preload`, API `max-age=15552000; includeSubDomains`)	#85 (ZAP report), #84
Safety	SAFE-1, SAFE-2	SAFE-1: server side passes on Render (refresh, reopen and lost-response retry keep every acknowledged answer and the original `endsAt`); the browser check passes on the local build in Chromium (TC-SAFE-01). SAFE-2: passes on the local build (TC-SAFE-02), not run on Render because deactivation cannot be undone	#96, section 4.5
Usability	QA-1	Not met yet: 1 of 1 verified participant applied within 3 minutes (P1, 2:33, Q1 = 4/5); the plan needs 5 students	#97 (uat-and-devices.md)
Availability	QA-2	Not measurable within the semester (Test Plan section 4)	
Reliability	QA-3	540 of 540 automated tests pass (487 server, 53 client); TC-REV-04 concurrent change passes on the deployed build	Unit Test Report, TC-REV-04
Maintainability	QA-4	Pass: server coverage 96.16% statements, 94.37% branches on the final commit (509 tests); client 39.85% branches (64 tests)	Branch Coverage Report section 7

Area	Requirement	Result	Source
Portability	QA-5	Not run as a device matrix: the eight rows in uat-and-devices.md section 6 are empty. Evidence so far: P1 completed S2 on a real Android phone in Chrome; the student flow passed in Chromium at 360 px on the local build (#96)	#97

7. User acceptance testing

Plan and script: uat-and-devices.md (#97). The session moved from 19–21 Nov into the test week (DEVIATIONS row 27).

Participant	Device	Task S2 (find and apply, target 3:00)	Q1
P1	Samsung phone, Chrome for Android, remote	2:33, completed without help	4 / 5

Only P1 is verified; P2–P5 and the recruiter R1 have not been recorded, and no usability defects were raised. QA-1 needs five students, so it is not met yet.

8. Defects

Issue	Title	Severity	Module	Found by	Status
#99	API fails to start in production: Cannot find module 'mongodb'	Critical (blocks deployment)	Identity (resume storage)	Deployment, 02-11-2026	Fixed in PR #100, merged and deployed 04-11-2026
#118	TC-TST-06: answer saves exceed 1 s at the client on the Render build (TST-4)	Low (no data loss)	Aptitude	#96, 05-11-2026	Recorded as DEVIATIONS row 28 (PR #119): mostly network latency to the free Render instance; local max 29 ms
#123	PERF-2: non-search API p95 3.6 s on the Render build	Medium (performance)	Listings / API	#89, 06-11-2026	Recorded as DEVIATIONS row 29: free Render tier and cross-region database; local p95 62 ms
#124	TC-SEC-02: login rate limit not enforced on the Render build	Medium (security)	Identity	Rerun, 06-11-2026	Recorded as DEVIATIONS row 30: rate-limit key changes behind the Render proxy; account lockout (TC-AUTH-03) still stops password guessing per account

Observations (not defects against the SRS):

• **O-1, email configuration.** When `SMTP_HOST` is not set, the mailer uses a JSON transport that sends nothing but reports success, so notifications show email status `SENT`. This is intended for development, but in production it hides a missing setting, as in the first applications run (section 4.2). A warning at startup when `NODE_ENV` is `production` and `SMTP_HOST` is empty would make this visible.

• **O-2, public listing count.** The deployed public listings API reported 4,421 records initially and 4,427 after Issue #88 test listings were created. Seed documentation describes 4,500 open listings. The difference has not been reconciled; investigate the seed, listing status and public-query criteria before treating it as a defect.

9. Metrics (Test Plan section 14)

Metric	Value
Test cases executed	51 of 51 (39 pass, 4 partial, 2 fail, 6 blocked); 0 not run
Test cases passed / failed	39 / 2 (full passes; partial cases excluded)
Blocked cases	6 (5 identity: admin or database access; PERF-3: dataset)
Defects by severity	Critical 1 (fixed), High 0, Medium 2 (#123, #124, deviations), Low 1 (#118, deviation)
Defects by module	Identity 2, Aptitude 1, Listings / API 1
Defect age (days open)	#99: 2 days (02-11 to 04-11); #118: 1 day (05-11 to 06-11); #123, #124: opened 06-11, open as deviations
Requirement coverage (share of IDs with a passed test)	100% with an automated test; see section 5 for deployed-build passes
Server statement coverage	96.16%

10. Conclusion and sign-off

Release 1.0 runs on the deployed build with no open Critical or High defect. All 51 test cases were executed: 39 pass, 4 partial, 2 fail and 6 blocked, all nine applications cases pass their API steps, the full Postman collection passes (157 assertions, 0 failures), SEC-2 passes, and server coverage is 96.16% statements and 94.37% branches with all 509 server and 64 client tests passing.

Every planned test case was executed. Six are blocked on Render because testers had no admin or database access (they pass in the automated suite) or no 2,000-student dataset (PERF-3, passes locally). Two fail on the free Render environment and are recorded as deviations: TC-TST-06 (#118, row 28) and TC-SEC-02 (#124, row 30); PERF-2 inside TC-PERF-01 also fails there (#123, row 29). UAT has one participant of five and the device matrix is not filled (#97). These gaps are recorded, not closed.

Role	Name	Date
QA lead	Guru R Bharadwaj	07-11-2026
Dev lead	E S Sukruth	
Members	Deekshith Gowda G, Sharath H G	